

Possibility of Age Determination on Otoliths, For Scad Mackerel, *Decapterus macarellus* and *Dobrada, Spicara melanurus* in the Cabo Verde waters

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Abstract

This study was made to examine possibility of age determination based on periodicity of the rings for otoliths for both Scad Mackerel (*Cavala preta*) and Dobrada in Cabo Verde waters. Biological information and otoliths were taken from a total of 411 of *Cavala preta*, and 682 of Dobrada around S. Vicente Island during November 2002 and October 2003. Transparency band was periodically formed, as observing posterior part of otolith, and was defined as annulus for both species. It was confirmed that burnt method was effective to enhance readability of annulus for both species. Applied on the burnt method, age determination of Dobrada was carried out, and 93% of otolith was successfully identified, using samples from February to May 2003. Von Bertalanffy growth parameters were obtained.

Introduction

Scad Mackerel, *Decapterus macarellus*, and Dobrada, *Spicara melanurus* are economically important species in Cabo Verde. Scad Mackerel and Dobrada are locally called *Cavala preta* and Dobrada, respectively. *Cavala preta* and Dobrada are caught mainly by purse seine and gillnet, respectively. Most of *Cavala preta* are landed in S. Vicente, and for Dobrada in Santiago Island.

Age information is the most important to see dynamic of fish population. In temperate or high latitude waters, age determination is a common practice, but in sub-tropical or tropical waters, periodic rings of hard parts, such as otoliths or scales, are usually obscured. Prior to age determination, periodicity of the ring formation on the hard parts should be confirmed.

The purpose of this study is to examine possibility of age determination based on periodicity of the rings for otoliths for both *Cavala preta* and Dobrada in Cabo Verde waters.

Material and Methods

Otoliths from both species, *Cavala preta* and Dobrada were collected for one year from November 2002 to October 2003. Samples for *Cavala preta* were obtained from R/V Sinagoga around S. Vicente Island. Samples of Dobrada, which were captured around S. Pedro in S. Vicente Island, were collected mainly from Mindelo fish market. Samples were taken to the INDP Laboratory. Length and weight were measured, sex was identified. Gonad stage by the visual assessment and gonad weight were recorded. After all measurements were made, a pair of largest otolith, sagittae, was removed. Removed otoliths were washed well by waters, and then put into small vial. Within one or two days, otoliths were observed.

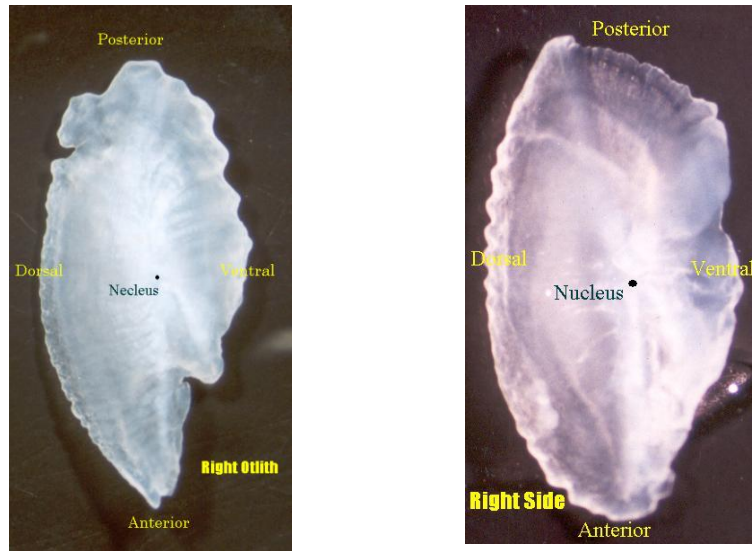


Figure 1. otolith structure of Cavala preta(Left) and Dobrada (right).

Otoliths were observed by the microscope under the black background with reflected light to judge transparency or opaque bands at the posterior margin of the otolith (Figure 1). Transparency band was observed as blackish color due to the black background, and opaque band was as whitish in reflection of color of otolith. Transparency band is usually formed when fish growth is slow, and this transparency band is defined as annulus for indication of age.

As reference, all otoliths were photographed. After photographing, otoliths were stored in a coin envelope for further process. Summary of collected otoliths samples were shown in Tables 1 and 2. Total of 411 of Cavala preta, and 682 of Dobrada were examined. If periodicity was confirmed, otoliths were further processed for establishment of method.

Table 1. Summary table of Cavala preta.

Year	Month	No	FL(mm)			
			Avg	Min	Max	Std
2002	11	44	209	187	288	21.36
	12	17	225	220	240	6.24
2003	1	20	248	210	280	20.42
	2	50	267	230	310	20.80
	3	50	269	236	304	14.69
	4	40	263	250	297	10.36
	5	55	249	163	305	48.12
	6	15	287	275	303	7.23
	7	20	282	262	316	11.78
	8	35	279	260	290	6.53
	9	35	241	178	305	40.07
	10	30	293	267	329	15.22641
Total		411				

Table 2. Summary table of Dobrada.

Year	Month	No	FL(mm)			
			Avg	Min	Max	Std
2002	11	60	206.6	182	235	14.02
	12	80	188.4	132	245	31.97
2003	1	60	219.1	196	254	10.20
	2	50	219.0	203	260	11.76
	3	76	221.0	189	255	14.52
	4	59	228.6	200	258	13.18
	5	69	206.4	73	259	48.95
	6	60	219.1	193	253	13.83
	7	58	229.1	195	275	14.48
	8	40	200.1	146	262	39.68
	9	40	242.9	226	258	9.72
	10	30	230.4	211	255	10.76
Total		682				

Results

For a total of 411 otoliths of Cavala preta and 682 of Dobrada, edge of otolith was examined whether opaque or transparency was formed. Obscured but observable transparency and opaque rings were alternately formed at the margin of the surface of otolith (Figure 1). Each otolith was observed, and assigned as transparency or opaque band. Monthly average ratio of appearance of the transparency bands was plotted in Figure 2.

Based on Figure 2, transparency and opaque bands were formed periodically for one-year period for both Cavala preta and Dobrada. Transparency band was formed between June and August for Cavala preta and between January and May for Dobrada. Thus, this transparency zone can be

defined as annulus, and July for *Cavala preta* and March for *Dobrada* are defined as a month of birth, respectively

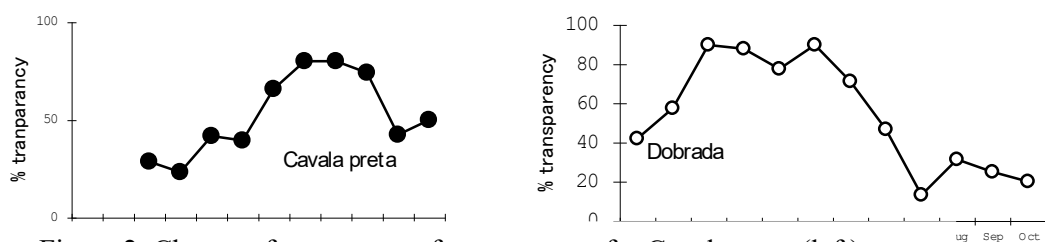


Figure 2. Change of percentage of transparency for *Cavala preta* (left) and *Dobrada* (right).

From observation of surface of otoliths for both *Cavala preta* and *Dobrada*, central part were quite difficult to discerned annulus. In this study, burnt method was employed. For October samples for both species, surface-burnt method (Christensen, 1964) and section-burnt method (Beamish and Chilton, 1982) were employed in order to examine readability of annulus.

For both surface-burnt and section-burnt methods for both species showed enhance of readability of annulus (Figures 3 and 4). Counts of annulus from both methods were the same and processing for the surface-burnt method was much easier than the section-burnt method, thus the surface-burnt method was recommended for age determination for *Cavala preta* and *Dobrada*.



Figure 3. Comparisons of readability of annulus between surface-burnt (left) and section-burnt (right) methods for *Cavala preta*.

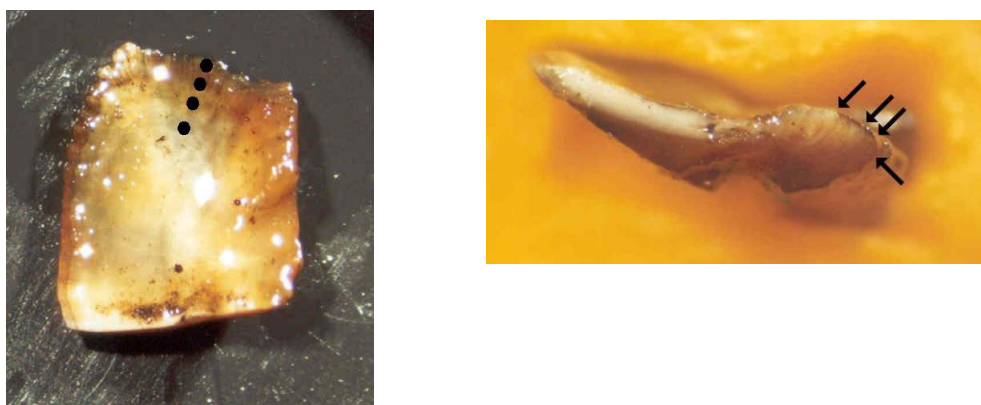


Figure 4. Comparisons of readability of annulus between surface-burnt (left) and section-burnt (right) methods for *Dobrada*.

Some otoliths were easily, but some of them were difficult. When compared with otolith of fish from temperate or high latitude, annulus identification was much difficult.

Using the surface-burnt method, age determination for Dobrada was carried out. A total of 265 otolith collected between February and May were examined. 248 otoliths were successfully determined, and rate of age readability was 93.6 %.

According to the results of age determination, maximum determined age was five.

Otolith radius, which was measured between nucleus and marginal edge, was linearly related to body length (Figure 5). The relationship between otolith radius (OR) and fork length (FL) was:

$$FL = 45.03 + 62.74 \times (OR) \quad (r = 0.794; n = 248)$$

Where r : correlation coefficient; and

n = number of otolith measured.

Thus, back calculation was made. Back calculated Fork lengths from age one to five were 12.9, 17.6, 20.3, 20.9, and 21.0 cm, respectively (Table 3). Using these age and length data, three parameters of Von Bertalanffy Growth Equation were estimated using the Walford Plot. For estimated parameters, $L_{\infty} = 21.4$ cm, $K = 0.853$ /year, and $t_0 = -0.183$. A fit of the growth curve was shown in Figure 6.

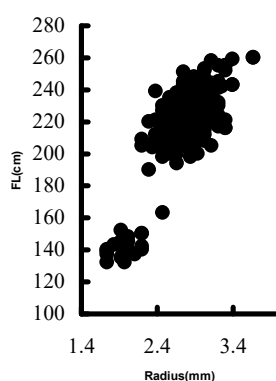


Figure 5. Relation ship between otolith radius and fork length (FL) for Dobrada.

Table 3. Back calculated length at age for Dobrada.

Age	AV.FL(cm)
1	12.89
2	17.62
3	20.25
4	20.85
5	20.99

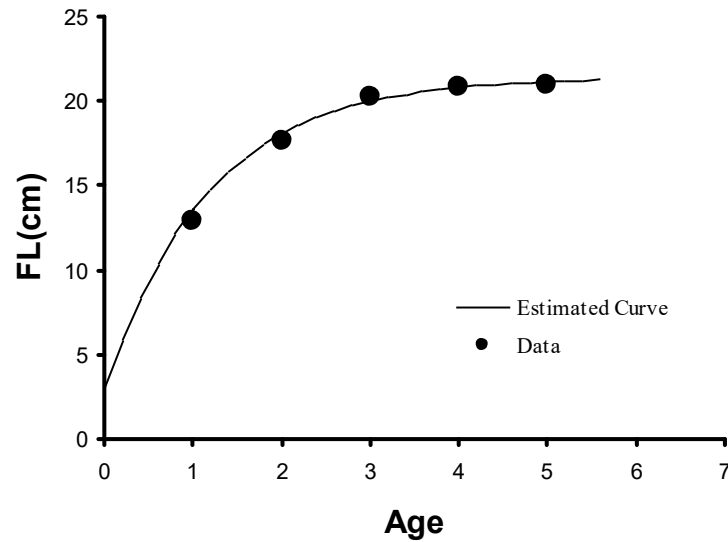


Figure 6. Estimated growth curve of Dobrada.

Discussions

This report examined that possibility of age determination for both Cavala preta and Dobrada, using otolith. Only one method could be employed in this report although Campana (2001) showed various methods for validation. Transparency and opaque bands were formed in one-year period. Transparent band can be defined as annulus. Although annulus in central part of otolith for both species were obscured, burnt method enhanced readability of annulus. Age determination could be possible for both species, using burnt method.

Annulus formation of hard parts including otolith was formed during slow fish growth period. Figure 7 showed changes of GSI (Gonad Somatic Index) of Dobrada from Costa (2004). During otolith samples collected from November 2002 to October 2003, timing of annulus formation and peak of GSI were coincident. However, patterns of changes of GSI between 2002 and 2003 were different. If peak of GSI, which may be correspondent to spawning period, influence formation of annulus of otolith, formation of annulus can be changed. This kind of seasonal changes can create difficulties of age reading of Dobrada.

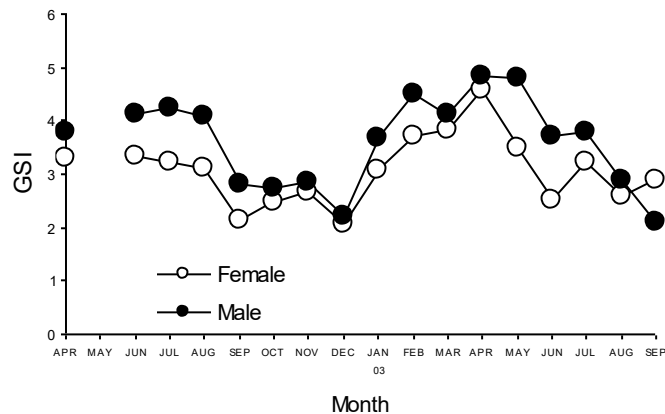


Figure 7. Changes of GSI for Dobrada from April 2002 to September 2003

Literature Cited

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